

Short-Horizon Trajectory Forecasting from GPS-Derived Motion Features Using Deep Sequence Models

A Comparative Evaluation of GRU, LSTM, and TCN Models for Short-Term Motion Prediction

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ABSTRACT

This project evaluates short-horizon trajectory forecasting using only GPS-derived motion features from the Microsoft GeoLife GPS Trajectory Dataset. The objective is to determine whether future displacement can be predicted one to five seconds ahead from latitude, longitude, and timestamp information alone, without relying on richer sensor inputs such as IMUs, cameras, or wheel-speed sensors. To address this question, the study compares GRU, LSTM, and Temporal Convolutional Network models against a constant-velocity baseline using engineered kinematic features derived after projecting GPS coordinates to UTM Zone 48 in meters.

The final pipeline drops malformed, duplicate, and non-forward timestamps; rejects GPS jumps greater than 100 meters; projects WGS84 latitude and longitude to UTM Zone 48; resamples trajectories to 1 Hz using time interpolation; and predicts future displacements rather than absolute coordinates. The six model features are dx, dy, speed, acceleration, heading, and turn rate. Model performance is evaluated with displacement-based metrics in meters, including ADE, FDE, and MDE. A key limitation is that global

normalization statistics were computed across the full dataset before splitting, so validation and test trajectories influenced the shared scaling parameters.

Results show that all learned models outperform the constant-velocity baseline, with the tuned LSTM delivering the strongest overall test performance among the evaluated architectures. The tuned LSTM achieved test ADE of 0.0020 m, FDE of 0.0032 m, and MDE of 0.0039 m, compared with a substantially larger constant-velocity baseline MDE of 8.448 m. Overall, the findings show that GPS-derived motion features can support accurate short-term displacement forecasting, while also highlighting the importance of transparent preprocessing assumptions, coordinate-system documentation, and normalization caveats when interpreting very small error values.

KEYWORDS

Short-horizon trajectory forecasting; GPS-derived motion features; GeoLife dataset; deep learning; GRU; LSTM; Temporal Convolutional Network; kinematic feature engineering; mobility analytics; real-time prediction.

1 Introduction

This project addresses the problem of short-horizon trajectory forecasting using only timestamps and GPS coordinates, with the goal of predicting an agent’s future displacement one to five seconds ahead. The central question guiding this work is: How accurately can we forecast near-term motion using only raw latitude, longitude, and time, without access to richer sensor suites such as IMUs, cameras, or wheel-speed sensors?

This problem is important because many operational systems must function in settings where sensor availability is limited, degraded, or inconsistent. A model capable of inferring short-term motion from sparse GPS feeds can support downstream tasks such as anomaly detection, collision risk estimation, and real-time decision support. In practical deployments, this capability can improve safety, reduce operational risk, and enhance situational awareness without requiring expensive or complex hardware.

The primary end users of this system depend on the deployment context. In operational environments, engineers and analysts may use trajectory forecasts to detect deviations from expected behavior or to support automated decision-making. In research or educational settings, the end user is the practitioner evaluating model performance and exploring the limits of GPS-only forecasting. In both cases, the value lies in producing accurate short-term displacement forecasts that could eventually be integrated into larger analytical or control pipelines.

This project uses the Microsoft GeoLife GPS Trajectory Dataset, a large-scale collection of real-world GPS trajectories recorded by 182 users over a five-year period. Each trajectory includes timestamped latitude, longitude, and altitude observations that support the derivation of motion

features such as speed, heading, acceleration, and turn rate after projecting coordinates into a meter-based planar reference frame. Because many trajectories are sampled at short intervals and contain realistic GPS noise across diverse movement patterns, the dataset is well suited for evaluating GPS-only short-horizon forecasting.

The overarching goal of this project is to build a lightweight forecasting pipeline capable of predicting future displacements over short horizons. The working hypothesis is that, even with limited inputs, derived kinematic features combined with sequence-based neural models such as LSTMs or temporal convolutional networks can achieve accurate and stable predictions. The final deliverable is an evaluated prototype pipeline that accepts sequences of GPS-derived motion features and outputs predicted displacements for the next several seconds. Model performance is evaluated using displacement metrics, training stability, and visual comparisons between predicted and actual trajectories.

2 Data Summary

The Microsoft Research Asia GeoLife GPS Trajectory Dataset version 1.3 provides time-ordered GPS traces collected from 182 users over a five-year period (Microsoft Research Asia, 2012). Each raw trajectory contains timestamped latitude, longitude, and altitude observations. For this project, the dataset is valuable because it combines realistic location noise, diverse movement patterns, and many trajectories sampled near 1 Hz, making it well suited for evaluating GPS-only forecasting over a one-to-five-second horizon.

Preprocessing converted raw GPS traces into physically interpretable motion sequences in meters. The final pipeline does not use an East-North-Up frame and does not operate directly in

standardized latitude/longitude space. Instead, WGS84 latitude and longitude are projected to UTM Zone 48, producing planar x and y coordinates in meters. The pipeline removes malformed records, duplicate timestamps, and non-forward time steps; rejects GPS jumps greater than 100 meters; resamples trajectories to 1 Hz using time interpolation; and generates prediction targets as future displacements relative to the last observed window position. Global normalization was applied before the trajectory-level split, and per-window normalization was added in the tuned pipeline; therefore, the shared scaling statistics should be treated as a methodological limitation rather than as evidence of a leakage-safe preprocessing design.

The main engineered features are dx, dy, speed, acceleration, heading, and turn rate. Position deltas are computed from projected meter coordinates, while speed and acceleration are derived from displacement and elapsed time. Heading is computed with $\arctan2$, and turn rate is calculated from heading differences divided by elapsed time. Because heading is not unwrapped at $\pm\pi$ in the current pipeline, discontinuities near the wrap boundary can produce artificial turn-rate spikes. This limitation is important for interpreting the acceleration and turn-rate distributions and motivates future angular preprocessing improvements.

Exploratory Data Analysis

Exploratory analysis was conducted on a filtered sample of 296 trajectories, yielding 364,187 motion records after quality filtering. The EDA applied the following quality filters:

- $0 < \Delta t \leq 60$ seconds
- $\text{speed} \leq 60$ m/s
- $|\text{acceleration}| \leq 20$ m/s²
- $|\text{turn rate}| \leq 5$ rad/s

Figure 1 summarizes the distribution of timestamp intervals between consecutive GPS observations, showing whether the filtered data are dense enough to support one-to-five-second forecasting. Most usable records occur at short sampling intervals, while occasional larger gaps identify irregular sampling that can introduce noise into derived speed, acceleration, and turn-rate features.

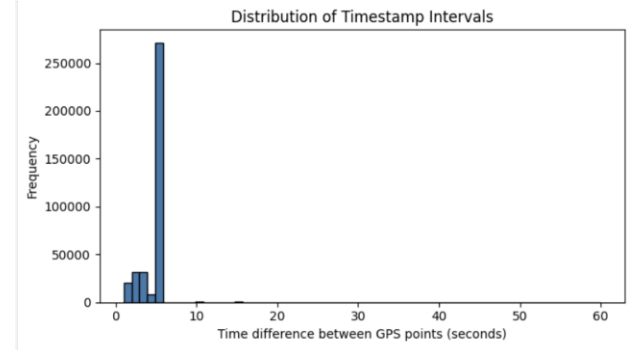


Figure 1. Distribution of timestamp intervals between consecutive GeoLife GPS observations. Most points occur at short intervals, while longer gaps identify irregular sampling that can introduce noise into derived motion features.

Figure 2 shows a representative speed profile after filtering and resampling. The profile illustrates that many short segments change smoothly across adjacent timestamps, supporting the use of sliding windows for near-term prediction. Isolated spikes still appear around irregular timestamp gaps or noisy GPS observations, suggesting that some extreme values reflect sampling artifacts rather than true movement changes. This pattern helps justify both the preprocessing filters and the inclusion of acceleration as a feature, since changes in speed provide useful short-horizon motion context.

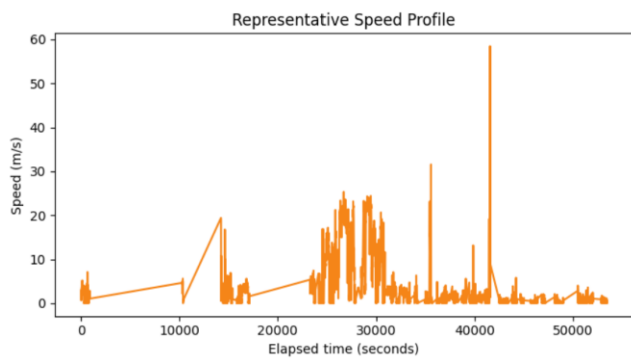


Figure 2. Representative speed profile from GeoLife trajectories. Speed generally changes smoothly over short horizons, with isolated spikes likely associated with timestamp irregularities or GPS noise.

Figure 3 shows the distributions of acceleration and turn rate after filtering. These variables provide additional evidence of temporal structure in the projected motion data: straight segments tend to produce stable headings and low turn rates, while turns produce larger rotational changes. The figure also highlights a known limitation of the current pipeline because heading is not unwrapped at $\pm\pi$ before differencing, which can create artificial turn-rate spikes near the wrap boundary. Overall, the distributions show that the GeoLife data contain meaningful near-term motion dynamics while also requiring careful filtering and angular preprocessing to manage irregular sampling and coordinate-derived artifacts.

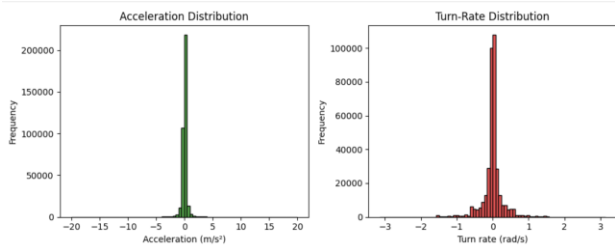


Figure 3. Distributions of acceleration and turn rate. Both variables are more volatile than speed and highlight the combined influence of real movement changes, irregular sampling, and heading discontinuities.

3 Literature Review

Short-horizon trajectory forecasting has been explored extensively across transportation analytics, mobility modeling, and geospatial prediction, with numerous methods designed to estimate near-term motion from positional data. Early approaches relied on classical kinematic models that assume constant velocity or constant acceleration. These models propagate future positions by integrating estimated speed and heading over short horizons and have been widely used in navigation and robotics applications due to their simplicity and low computational cost (Teunissen & Montenbruck, 2017). Although effective in stable motion regimes, these approaches struggle with nonlinear trajectories, sharp turns, and GPS noise, limiting their accuracy in real-world environments.

More sophisticated forecasting systems incorporate Bayesian filtering techniques such as the Kalman filter and its variants. These methods model motion as a dynamical system with latent velocity and acceleration states, enabling smoother estimates of position and heading and reducing the impact of GPS jitter. Kalman-based approaches have been applied extensively in vehicle tracking, pedestrian localization, and sensor fusion tasks, demonstrating improved stability over purely kinematic models (Sola, 2017). However, these methods often assume linear dynamics or require additional sensors such as IMUs to achieve high accuracy, making them less suitable for GPS-only scenarios.

Commercial mobility platforms also address short-term motion prediction using map-matching algorithms, route-constrained forecasting, and learned travel-time models. Ride-sharing and navigation systems frequently combine GPS traces with road-network information to infer likely paths and estimate near-term movement.

Large-scale navigation systems such as Google Maps demonstrate the practical importance of learned mobility models for estimating arrival times and detecting deviations from expected behavior (Derrow-Pinion et al., 2021).

Recent advances in machine learning have led to the adoption of sequence-based neural models for trajectory prediction. Recurrent neural networks, particularly long short-term memory (LSTM) networks and gated recurrent units (GRUs), were originally developed to capture long-range temporal dependencies in sequential data (Hochreiter & Schmidhuber, 1997; Cho et al., 2014). These architectures have since been adapted for mobility forecasting, where they can learn nonlinear relationships between displacement, speed, heading, and future motion. The GeoLife dataset itself has also been widely used to study human mobility patterns from GPS traces (Zheng et al., 2008), and Zheng et al. (2010) later described GeoLife as a collaborative social networking service connecting users, locations, and trajectories.

Temporal convolutional networks (TCNs) represent another major development in sequence modeling. By using dilated causal convolutions, TCNs can model long temporal contexts efficiently while maintaining stable gradients and low inference latency. Bai et al. (2018) demonstrated that TCNs outperform many recurrent architectures on sequence-prediction tasks, making them promising for real-time forecasting applications. Their ability to process sequences in parallel also makes them attractive for deployment in systems requiring rapid inference.

Several academic and commercial projects illustrate methods related to the problem addressed in this capstone. Recurrent models such as LSTMs and GRUs are commonly used for

sequential prediction problems because they can model temporal dependencies in ordered data, while TCNs have been used in real-time sequence modeling contexts where low latency and temporal stability are important (Bai et al., 2018). Commercial navigation systems combine kinematic, map-based, and learned models to estimate travel time and detect anomalous movement. Collectively, these examples demonstrate that sequence-based neural architectures and derived kinematic features are relevant to GPS-only short-horizon trajectory forecasting.

4 Methodology

The forecasting system developed for this project uses engineered kinematic features, sliding-window sequence extraction, and three deep learning architectures—GRU, LSTM, and a Temporal Convolutional Network (TCN)—to predict short-horizon displacement from GPS-derived inputs. A classical constant-velocity baseline model is included to contextualize the performance of the learned models. This section describes the model architectures, preprocessing pipeline, training procedures, and optimization strategy used in the project.

The forecasting problem is formulated as a supervised sequence-to-sequence task. Each model receives a fixed-length input window of twenty consecutive motion states and predicts two-dimensional displacement for the next five time steps. Raw latitude and longitude are first projected from WGS84 coordinates into UTM Zone 48 so that model inputs and displacement targets are represented in planar meters. Each motion state includes six engineered features: dx , dy , speed, acceleration, heading, and turn rate. These variables capture translational and rotational motion dynamics from GPS-only inputs

while keeping the forecasting horizon limited to one through five seconds ahead.

A simple kinematic baseline model was implemented to provide a lower-bound reference for evaluating the learned architectures. The baseline assumes constant velocity over the prediction horizon by extrapolating the most recent observed displacement. It computes the difference between the final two positions in the input window and repeats this displacement for each of the five future timesteps. Using the same window generator and evaluation metrics as the learned models ensures a fair comparison.

Three neural architectures were developed to explore different approaches to temporal modeling. The GRU model consists of a single gated recurrent unit layer with 64 hidden units, followed by a fully connected layer that maps the final hidden state to the ten displacement outputs. The LSTM model mirrors this structure, replacing the GRU layer with a long short-term memory layer of the same width. Both architectures are designed to capture temporal dependencies through gating mechanisms that regulate information flow.

The TCN model incorporates a deeper and more expressive architecture. It uses five temporal convolutional blocks with dilation factors of 1, 2, 4, 8, and 16, enabling the receptive field to span and exceed the twenty-step input window. Each block includes a dilated convolution, batch normalization, ReLU activation, dropout, and a residual connection. Sinusoidal positional encodings are added to the input sequence to provide temporal context, and a multi-head self-attention layer with four heads processes the temporal features after the convolutional stack. A final fully connected layer maps the attended representation to the displacement outputs. This architecture was selected to evaluate whether

convolutional temporal modeling and attention mechanisms can outperform recurrent baselines in short-horizon forecasting.

Training follows a unified pipeline across all models. The GeoLife dataset is split at the trajectory level into 60 percent training, 20 percent validation, and 20 percent test sets to prevent segments of the same trajectory from appearing across splits. Before modeling, malformed rows, duplicate timestamps, and non-forward time steps are removed; GPS jumps greater than 100 meters are rejected; projected trajectories are resampled to 1 Hz using time interpolation; and future targets are encoded as displacements relative to the last observed window position. Global normalization is computed before the split and retained in the tuned pipeline, which means validation and test trajectories share the same scaling statistics as the full dataset. This is a methodological limitation and should not be characterized as leakage-safe. The tuned pipeline also applies per-window normalization to improve numerical stability, while a streaming generator creates input windows and displacement targets without loading all windows into memory simultaneously.

All models are trained using the Adam optimizer with a learning rate of 1×10^{-3} and mean squared error loss (Kingma & Ba, 2015). Training proceeds for twenty epochs, with progress and system-resource heartbeats logged periodically. Because some GeoLife trajectories contain tens of thousands of samples, adaptive caps limit the number of windows extracted from any single trajectory. The default cap is 5,000 windows, with lower caps for very large trajectory collections and model-specific overrides used during tuning. This approach maintains computational feasibility while preserving temporal diversity across the dataset.

Model optimization involved iterative adjustments to architectural and training hyperparameters. Hidden sizes of 32, 64, and 128 were evaluated for recurrent models, along with variations in dropout rates and learning rates. The tuned TCN uses a substantially deeper architecture than the pre-tuning version, including five dilated convolutional levels, dropout, batch normalization, residual connections, sinusoidal positional encoding, and four-head self-attention. Window caps were tuned to balance memory constraints with training diversity, including separate overrides for the recurrent models and the TCN. These optimization experiments informed the final architecture selections described above; comparative performance outcomes are presented in the Results section.

5 Results

Model performance was evaluated using Average Displacement Error (ADE), Final Displacement Error (FDE), and Maximum Displacement Error (MDE), three standard metrics for short-horizon trajectory forecasting. These metrics quantify how far predicted positions deviate from ground-truth future positions over the five-step prediction horizon. Because raw WGS84 latitude and longitude were projected to UTM Zone 48 before feature generation and modeling, the reported errors are interpreted in meters rather than degrees, standardized coordinate units, or normalized units. Evaluation was conducted on held-out test trajectories using displacement targets measured relative to the last observed position in each input window.

The corrected test results show that the tuned LSTM was the strongest overall model. It achieved test ADE of 0.0020 m, FDE of 0.0032 m, and MDE of 0.0039 m. The GRU was also highly accurate, with ADE of 0.0031 m, FDE of 0.0046 m, and MDE of 0.0052 m. The TCN achieved the

highest ADE among the learned models at 0.0033 m, indicating that its deeper convolutional and attention-based architecture did not surpass the recurrent models on this task. Across all learned models, error increased from ADE to FDE and MDE, showing that later prediction steps are generally more difficult than earlier ones over the one-to-five-second horizon.

The constant-velocity baseline provides an important contrast to the learned models. Its test MDE was 8.448 m, substantially larger than the learned-model errors. This gap confirms that simple extrapolation from the final observed displacement is not sufficient for noisy GeoLife GPS trajectories, even over a short five-second forecast horizon. The learned models were able to use recent motion history, speed, acceleration, heading, and turn-rate information to produce much smaller displacement errors.

Before comparing final test errors, training behavior provides important context for whether each architecture learned the displacement-prediction objective consistently. Figure 4 presents the training-loss curves for the GRU, LSTM, and TCN models during tuning. These curves show how quickly each architecture reduced mean squared error and whether optimization remained stable across epochs. The recurrent models converged smoothly, while the tuned TCN benefited from architectural changes and window-generation adjustments that improved stability compared with the original TCN implementation. Training loss is not the final evaluation criterion, but it helps explain whether each model learned consistently before being assessed on held-out test trajectories.

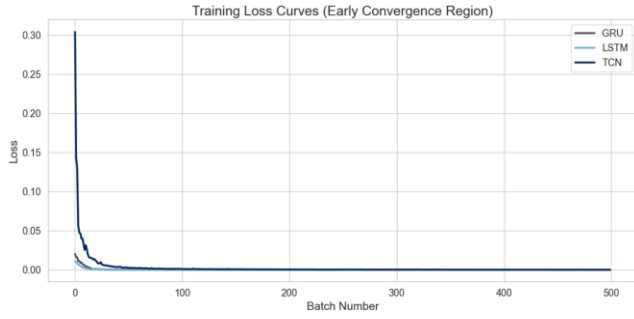


Figure 4. Training-loss curves for the GRU, LSTM, and TCN models during tuning. Lower loss indicates improved fit to the displacement-prediction objective.

Figure 5 expands the test-set evaluation by showing error distributions for all learned models. Unlike the mean ADE, FDE, and MDE values, these distributions show how displacement errors vary across individual prediction windows. This view helps distinguish models with consistently low errors from models whose average performance may hide occasional larger deviations. The LSTM shows the strongest overall test behavior, consistent with its lower ADE, FDE, and MDE values, while the GRU remains competitive and the TCN shows slightly weaker displacement-error performance.



Figure 5. Error distributions for all models on the test set. The distributions compare how displacement errors vary across prediction windows for the GRU, LSTM, and TCN models.

Figure 6 compares predicted versus true final displacements on the held-out test set. These plots diagnose whether each model's fifth-step predictions remain centered near the ground-truth

displacement targets or drift systematically away from them. The GRU and LSTM predictions cluster tightly around the true final displacements, with the LSTM producing the most compact pattern. This visual result supports the numerical finding that the tuned LSTM generated the most stable and accurate short-term displacement forecasts.

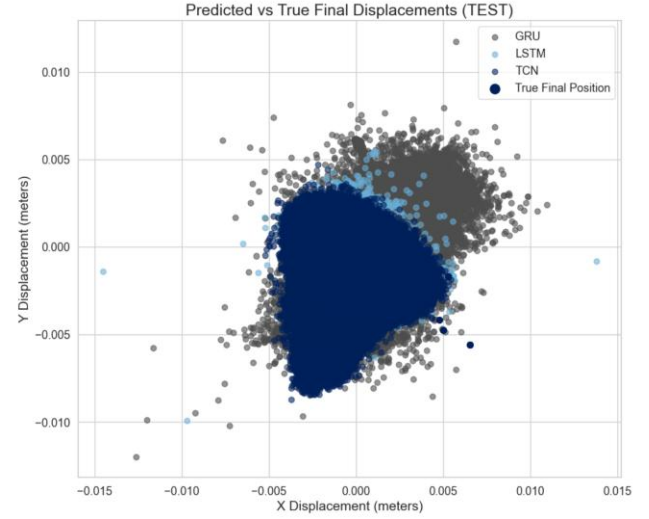


Figure 6. Predicted vs true final displacements on the test set. The plots compare each model's final-step displacement predictions with the corresponding ground-truth displacement targets.

Figure 7 serves as the final model performance dashboard for the held-out test set. It consolidates ADE, FDE, and MDE for the GRU, LSTM, and TCN models, making the relative strengths of each architecture easy to compare after reviewing the training curves, error distributions, and final-displacement diagnostics. The LSTM achieves the lowest errors across all three metrics, reflecting its strong temporal stability and consistent short-horizon accuracy. The GRU performs slightly worse but remains competitive, while the TCN has the highest ADE among the learned models. This dashboard reinforces the conclusion that the LSTM provides the most reliable overall displacement forecasts for this GPS-only task.



Figure 7. Model performance dashboard showing ADE, FDE, and MDE for the GRU, LSTM, and TCN models on the test set.

6 Conclusion

The goal of this project was to evaluate whether short-horizon displacement can be accurately forecast using only GPS-derived features—dx, dy, speed, acceleration, heading, and turn rate—without relying on richer sensor suites. The working hypothesis was that sequence-based neural models would outperform a simple constant-velocity baseline and produce stable predictions relevant to future real-time mobility applications. The results support this hypothesis. All learned models substantially outperformed the baseline, and the tuned LSTM achieved the strongest overall displacement-error performance on the held-out test set, with errors interpreted in meters because coordinates were projected to UTM Zone 48 before modeling.

The project also demonstrated that transparent preprocessing is central to interpreting model performance. Projecting WGS84 latitude and longitude to UTM Zone 48 made the displacement targets physically meaningful in meters, while resampling to 1 Hz and rejecting large GPS jumps reduced instability from irregular traces. Adaptive window caps, per-window normalization, and improved handling of long trajectories were especially important for stabilizing training in the

tuned pipeline. At the same time, global normalization statistics were computed across the full dataset before splitting, so scaling information was shared with validation and test trajectories. This should be treated as a methodological limitation rather than described as leakage-safe.

The results highlighted the sensitivity of short-horizon forecasting to irregular sampling intervals, GPS noise, and angular discontinuities in heading. Because heading is not currently unwrapped at $\pm\pi$ before computing turn rate, wraparound points can create artificial turn-rate spikes and inflate derived motion features. Future work should incorporate heading unwrapping, independent verification of the projected meter-based coordinate conversion on held-out traces, and more robust smoothing or interpolation strategies. These improvements would likely enhance model performance, especially in trajectories with inconsistent sampling rates or abrupt coordinate artifacts.

A valuable extension of this work would be to examine how different GPS sampling intervals affect model accuracy. The GeoLife dataset contains many trajectories sampled near one hertz, but real-world systems often operate at lower or variable rates due to device limitations, power constraints, or environmental conditions. Systematically degrading the sampling frequency and evaluating the resulting impact on ADE, FDE, and MDE would reveal the minimum viable sampling rate at which the learned models maintain reliable short-term forecasts. This analysis would provide practical guidance for deployment scenarios where GPS updates may arrive more slowly and would help determine whether additional temporal modeling or interpolation strategies are required to preserve accuracy under reduced sampling conditions.

Several additional extensions would strengthen the forecasting system and broaden its applicability. Transformer-based architectures represent a promising direction because self-attention mechanisms can model longer temporal dependencies and may better adapt to irregular sampling when paired with appropriate preprocessing (Vaswani et al., 2017). Evaluating performance across different transportation modes such as walking, biking, and driving would provide deeper insight into how the models behave under distinct motion regimes. Developing a production-ready inference module with real-time streaming, latency profiling, and monitoring would enable deployment in operational environments where rapid and reliable short-term motion prediction is required.

Overall, the results demonstrate that accurate short-horizon displacement forecasting is feasible using GPS-derived features and lightweight temporal models. The final pipeline provides a strong foundation for future work in real-time mobility analytics while emphasizing the importance of coordinate-system documentation, meter-based error interpretation, normalization transparency, sampling-rate analysis, and angular-feature handling when working with large and heterogeneous geospatial datasets.

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